

Level 3 - Searching Problems

1. Search in Rotated Sorted Array

```
nums = [4, 5, 6, 7, 0, 1, 2]
target = 0

def search_rotated(nums):

    left = 0
    right = len(nums) - 1

    while left <= right:

        mid = (left + right) // 2

        if nums[mid] == target:
            return mid

        # Left half is sorted
        if nums[left] <= nums[mid]:

            if nums[left] <= target < nums[mid]:
                right = mid - 1
            else:
                left = mid + 1

        # Right half is sorted
        else:

            if nums[mid] < target <= nums[right]:
                left = mid + 1
            else:
                right = mid - 1

    return -1

print(search_rotated(nums))
```

```
def ship_within_days(weights, days):

    left = max(weights)
    right = sum(weights)

    answer = right

    while left <= right:

        capacity = (left + right) // 2

        if can_ship(weights, days, capacity):

            answer = capacity
            right = capacity - 1

        else:

            left = capacity + 1

    return answer

weights = [1, 2, 3, 1, 1]
days = 4

print(ship_within_days(weights, days))
```

2. Binary Search on Answer

```
def can_ship(weights, days, capacity):

    days_used = 1
    current_load = 0

    for weight in weights:

        if current_load + weight > capacity:

            days_used += 1
            current_load = weight

        else:

            current_load += weight

    return days_used <= days
```